

HR ANALYTICS PROJECT REPORT

Predicting Employee Attrition

Exploratory analysis and machine learning on the IBM HR Analytics dataset

Prepared by

Dilsha A

Tools

Python (pandas, scikit-learn, SHAP), VS Code, Power BI

Date

October 2026

Abstract

Employee attrition is a major cost for organisations because hiring and training replacements takes time and money. This project analyses the IBM HR Analytics dataset of 1,470 employees to identify the main factors linked to resignation and to build a model that predicts who is likely to leave. Exploratory analysis showed that overtime, lower monthly income, younger age and department are clearly associated with attrition. A logistic regression model reached 86% accuracy on unseen data, but recall for employees who leave was only 0.34 because just 16% of employees in the data are leavers. SHAP analysis confirmed overtime, frequent business travel, short working experience and long gaps since promotion as key drivers. The findings show that accuracy alone can be misleading on imbalanced data and that class balancing is the most useful next step.

1,470 Employees analysed	16% Attrition rate	86% Model accuracy	0.34 Recall (leavers)
------------------------------------	------------------------------	------------------------------	---------------------------------

1. Introduction

Employee attrition is costly for companies because hiring and training new staff takes time and money. This project analyses the IBM HR Analytics dataset of 1,470 employees to find the main reasons employees leave. It also builds a machine learning model to predict who is likely to resign.

The objectives are to: (1) describe how attrition varies across departments, overtime, income and age; (2) train and evaluate a model that predicts attrition; and (3) explain which factors influence the model most, so that HR teams can act on them.

2. Dataset Description

The IBM HR Analytics Employee Attrition dataset (WA_Fn-UseC_-HR-Employee-Attrition.csv) contains 1,470 employee records. Each record includes details such as age, department, job role, monthly income, overtime status, business travel, job satisfaction and years at the company. The target column is Attrition (Yes/No). About 16% of employees have left and 84% have stayed, so the classes are imbalanced. This matters when judging model performance.

3. Methodology

- **Data loading and checking.** The CSV was loaded with pandas and checked for missing values and data types.
- **Exploratory analysis.** Attrition was compared across department, overtime, monthly income and age using charts.
- **Dashboard data.** A cleaned file (hr_data_for_powerbi.csv) was exported for dashboarding in Power BI.

- **Preparation.** Categorical columns were one-hot encoded and numeric features were standardised with a scaler. The data was split into 1,176 training and 294 test records (80/20).
- **Modelling.** A logistic regression model (max_iter = 1000) was trained on the scaled training data.
- **Evaluation.** Performance was measured with accuracy, precision, recall, F1-score and a confusion matrix. SHAP values were used to explain feature impact.

4. Exploratory Analysis

4.1 Attrition by department

Sales has the highest attrition rate at 20.6%, followed by Human Resources at 19.0%. Research & Development has the lowest rate at 13.8%.

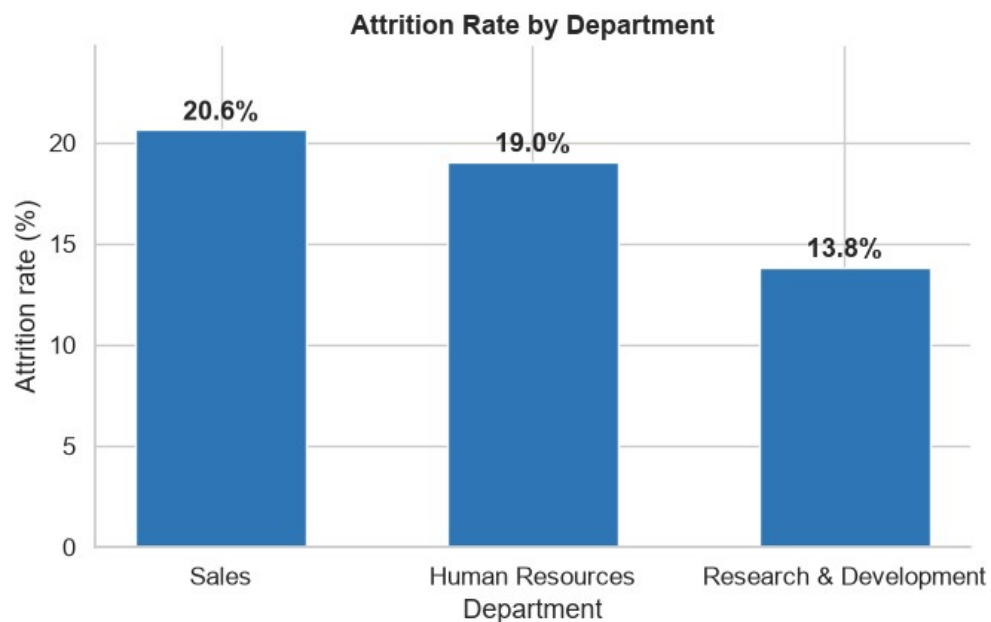


Figure 1: Attrition rate (%) by department

4.2 Attrition by overtime

Overtime is the clearest signal in the data. 30.5% of employees who work overtime leave, compared with 10.4% of those who do not, roughly three times higher.

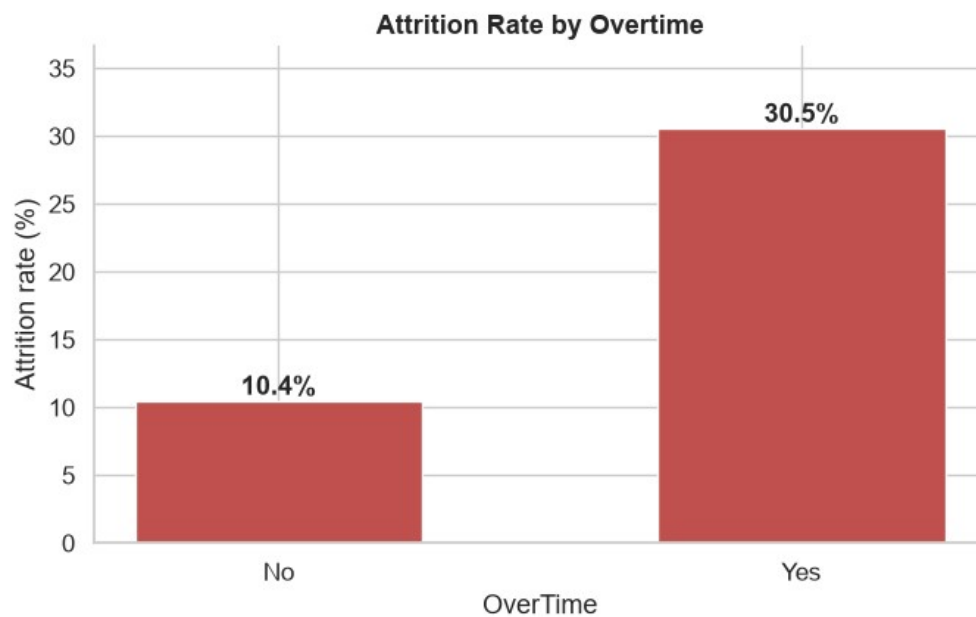


Figure 2: Attrition rate (%) by overtime

4.3 Monthly income

Employees who left have lower pay. Their median monthly income is about 3,200, compared with about 5,200 for those who stayed, and the middle half of leavers earn less than the middle half of stayers.

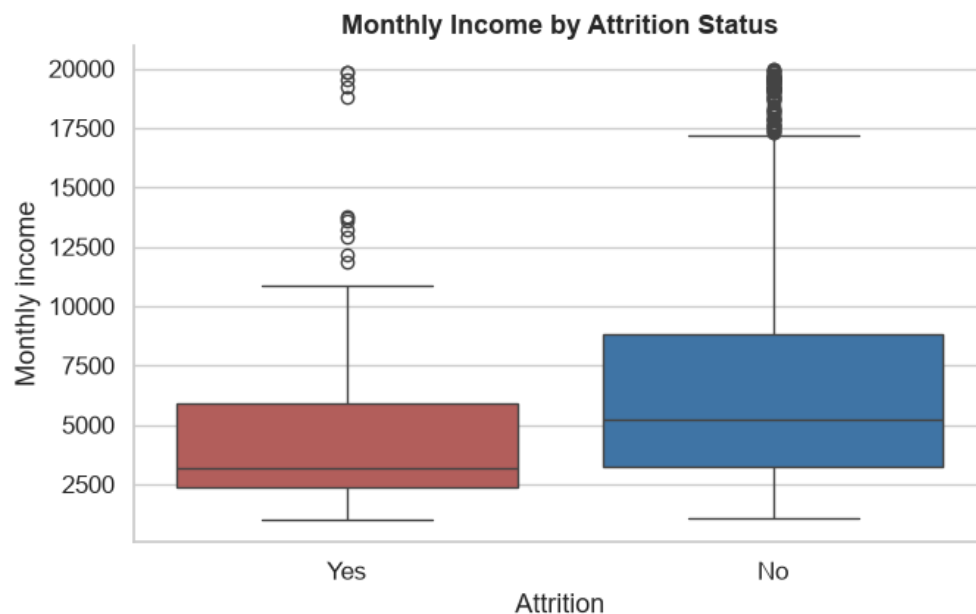


Figure 3: Monthly income by attrition status

4.4 Age

Leavers are concentrated among younger employees, mainly in their 20s and 30s. The share of leavers falls steadily after about age 40.

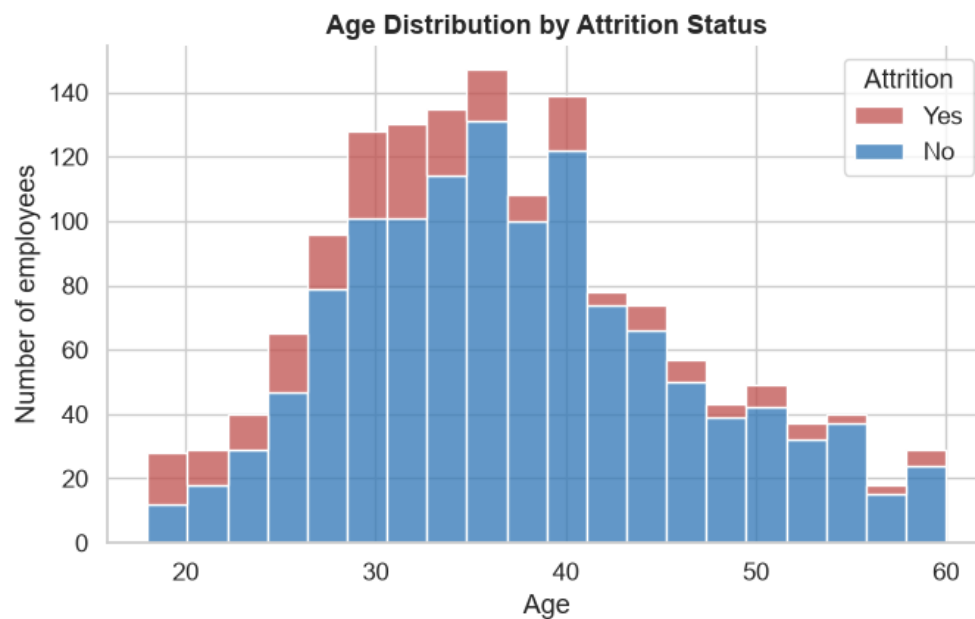


Figure 4: Age distribution by attrition status

5. Model Results

The logistic regression model achieved an accuracy of 0.86 (86%) on the 294 test records. The classification report is shown below.

Class	Precision	Recall	F1-score	Support
0 (Stayed)	0.88	0.96	0.92	247
1 (Left)	0.62	0.34	0.44	47
Accuracy			0.86	294
Macro average	0.75	0.65	0.68	294
Weighted average	0.84	0.86	0.84	294

Table 1: Classification report on the test set

The model is very good at identifying employees who stay (recall 0.96). For employees who leave (class 1), recall is only 0.34, so the model catches about 1 in 3 real leavers. When it predicts that someone will leave, it is correct 62% of the time (precision 0.62).

5.1 Confusion matrix

	Predicted: Stay	Predicted: Leave
Actual: Stay	237 (true negative)	10 (false positive)
Actual: Leave	31 (false negative)	16 (true positive)

Table 2: Confusion matrix on the test set

Of the 47 employees who actually left, the model found 16 and missed 31. It wrongly flagged only 10 of the 247 employees who stayed. The high accuracy therefore comes mainly from the large number of correct "stay" predictions.

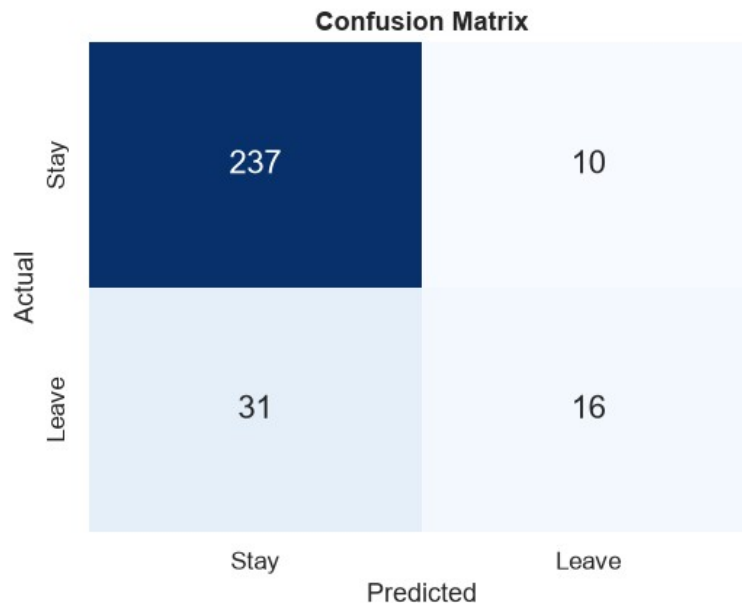


Figure 5: Confusion matrix (model output)

6. Feature Importance (SHAP)

SHAP values show how much each feature pushes the model towards predicting attrition. In Figure 6, red points are high feature values and blue points are low values; points to the right increase the predicted chance of leaving.

- **Overtime (OverTime_Yes):** Employees who work overtime are pushed strongly towards leaving.
- **Frequent business travel and Laboratory Technician role:** Frequent travellers and laboratory technicians show higher attrition risk.
- **Experience and promotion:** Fewer total working years and more years since the last promotion increase risk.
- **Satisfaction:** Higher environment and job satisfaction reduce the risk of leaving.
- **Marital status and job history:** Single employees and those who have worked at many companies are more likely to leave.

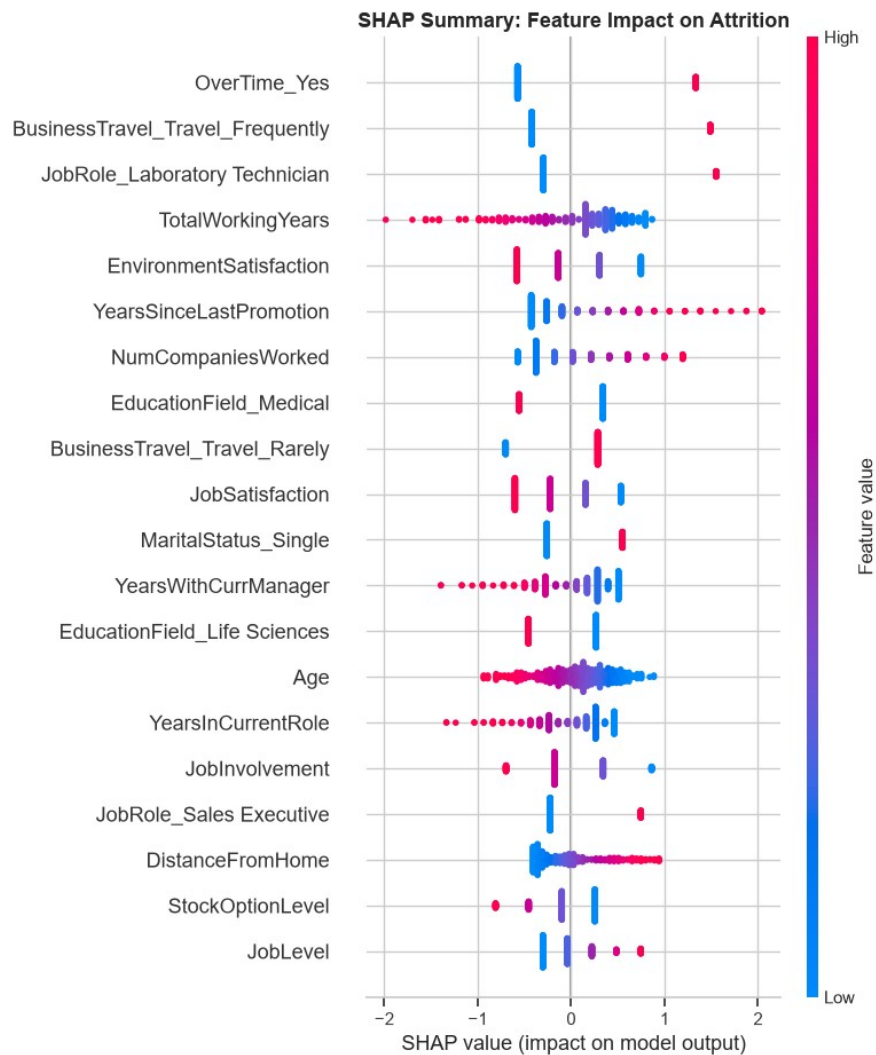


Figure 6: SHAP summary plot of feature impact

7. Conclusion

This project found that overtime, lower income, younger age and department are linked to employee attrition, and that a logistic regression model can reach 86% accuracy. However, the recall of 0.34 for leavers shows the model misses most employees who actually resign. Because the data is imbalanced, with only 16% leavers, accuracy alone overstates how good the model is.

Limitations and next steps

- Balance the classes with SMOTE, class weights or oversampling to improve recall.
- Tune the decision threshold to trade some precision for higher recall.
- Compare other algorithms, such as Random Forest or Gradient Boosting, using cross-validation.
- Note that the dataset is a single snapshot, so the results show association, not proof of cause.

8. Recommendations for HR

- **Reduce overtime.** Review workloads for employees working regular overtime, as this is the strongest driver of attrition.
- **Review compensation.** Check pay levels for lower-income and early-career staff, who are at higher risk.
- **Speed up promotion.** Build clear promotion paths, since long gaps since the last promotion raise risk.
- **Target high-risk groups.** Prioritise retention efforts in Sales and Human Resources and for younger employees.
- **Use the model carefully.** Use the model as an early-warning tool alongside manager feedback, not as the only decision maker.